

Design Details Document

Autonomous Medical Supply and Quarantine Robot

Robotics for Good Youth Challenge India 2026

Team code: ST-77

1. Objective and approach

The challenge represents a public health emergency. Our robot handles two jobs. It delivers ten medical kits in groups of 2, 6, and 2. It also carries and releases two wooden beams for the quarantine zone. These tasks move supplies and barriers without a person entering the field during the round.

The robot starts completely inside the official 480×280 mm starting zone. A team member switches it on and presses one button. From that point, the ESP32 runs the full route. The robot uses an ultrasonic sensor for wall distance, two downward-facing IR sensors for a black tape checkpoint, and calibrated turn timing. It stops after the final beam release.

One competition program: The submitted robot uses `MinimalistCareBot.ino`. The other sketches in the repository are kept for hardware checks, fault testing, and development. They are not extra operating modes for the competition robot.

Tasks completed by the program

1. Drive to the first wall checkpoint, turn left, and release two kits.
2. Detect the black tape checkpoint and release six kits.
3. Drive to the next wall checkpoint and release the other two kits.
4. Turn toward the quarantine area and stop at the next wall checkpoint.
5. Release the first beam, turn left, release the second beam, and stop.

The code releases complete preloaded groups. It does not count individual kits. This keeps the mechanism basic and reduces the number of moving parts.

2. Physical robot

The official robot is the four-wheel machine shown in the photographs on the next page. It has a flat wooden chassis, four rubber wheels, three open kit compartments, several wooden supports, and servo-operated mechanisms. The front ultrasonic sensor is visible on its wooden bracket. Electronics and wiring sit in the centre of the chassis.

The photographs show four geared drive units, one at each wheel. The program still controls two drive sides. The two left motors act as one left-side pair, and the two right motors act as one right-side pair. The final motor driver must safely handle the combined startup and stall current of each pair.

Three visible top-mounted servos operate the kit sections. Two more servo channels are reserved for the beam-release mechanisms. The supplied images do not show the downward IR sensors or every beam linkage clearly, so their final brackets must be checked on the actual robot before the run.

Size and mass. The supplied photographs do not provide a reliable scale or mass reading. The complete robot, including loaded kits and beams, is checked as one assembly against the 480×280 mm starting zone.

3. Official images of the built robot

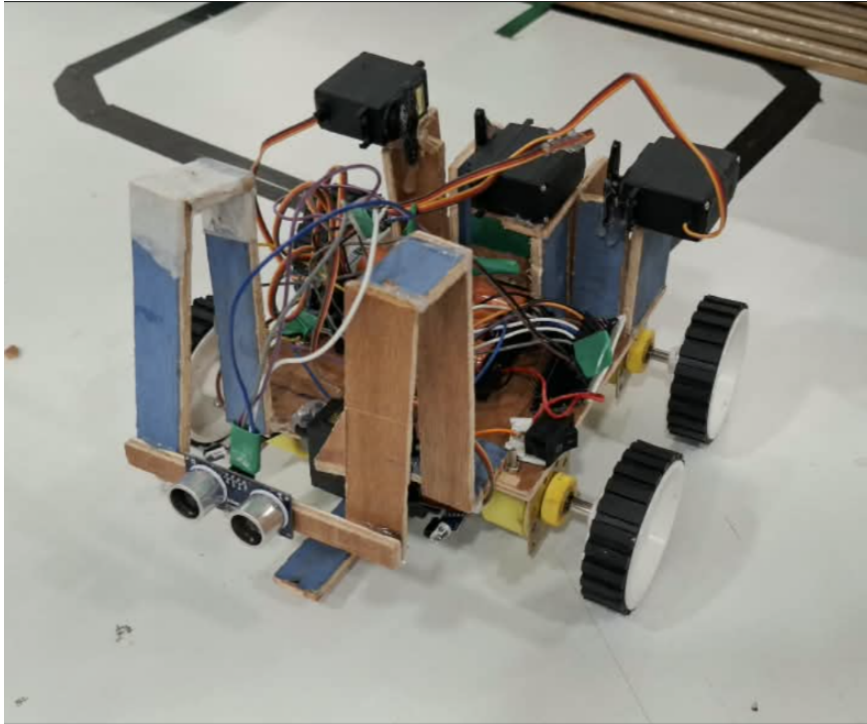


Figure 1. Front three-quarter view. The ultrasonic sensor, four-wheel base, kit compartments, and servo mounts are visible.

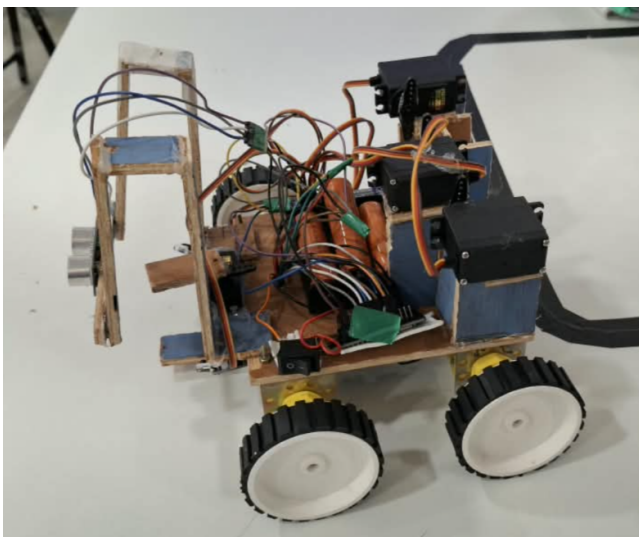


Figure 2. Side and rear structure.

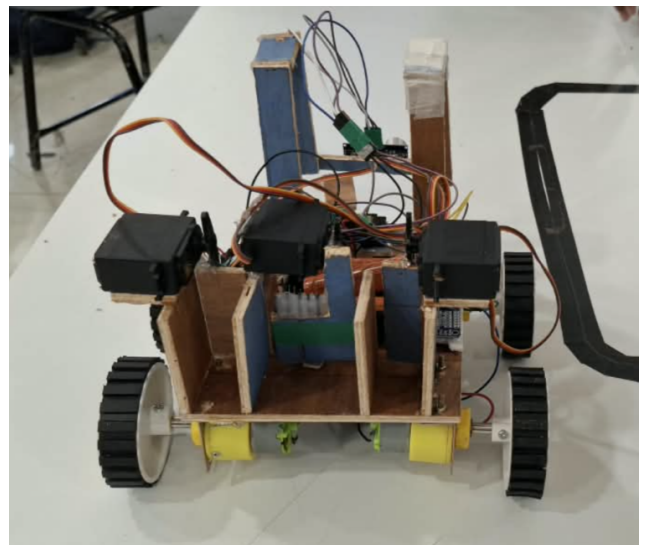


Figure 3. Four-wheel drive and three bin sections.

Model check

The Blender model follows the same main arrangement: four wheels, a low rectangular chassis, three delivery sections, and tall supports for the release mechanisms. The real photographs are the final reference. If the Blender file and the built robot differ, the built robot shown above takes priority.

4. Mechanism, electronics, and wiring

Part	Qty.	Purpose
ESP32-WROOM-32 DevKit	1	Runs the autonomous route and reads the sensors.
L298N dual motor driver	1	Controls the left and right motor pairs. Its current rating must be checked.
12 V geared DC motor	4	One motor at each wheel; two motors work together on each side.
PCA9685 servo driver	1	Sends 50 Hz commands to five servos over I ² C.
Positional servo	5	Three kit releases and two beam releases.
Ultrasonic sensor	1	Measures the distance to a wall in front of the robot.
Digital IR sensor	2	Detects a black tape stripe under the robot.
Start pushbutton	1	Starts one autonomous run.
Main power switch	1	Provides an accessible physical stop.

Kit and beam mechanisms

The three bins are loaded with 2, 6, and 2 wooden kits. Each kit is $25 \times 25 \times 20$ mm. Servo channels 0, 1, and 2 open the three release mechanisms. Gravity drops each complete group while the robot is stopped. Two holders carry the quarantine beams. Each beam has a 60×20 mm section. Servo channel 3 releases the first beam, and channel 4 releases the second. The program turns left between the two releases. The holder positions must allow each beam to leave the robot cleanly and stand in the required location.

ESP32 and PCA9685 connections

GPIO / channel	Connection	Job
25, 26	L298N IN1, IN2	Left-side motor-pair direction
27, 14	L298N IN3, IN4	Right-side motor-pair direction
33, 17	L298N ENA, ENB	Left and right PWM speed
23, 34	Ultrasonic TRIG, ECHO	Wall-distance measurement
35, 16	Left and right IR outputs	Black tape detection
21, 22	PCA9685 SDA, SCL	Servo-driver I ² C link
32	Pushbutton to ground	Start input with internal pull-up
PCA 0, 1, 2	Three kit servos	2-kit, 6-kit, and 2-kit release
PCA 3, 4	Two beam servos	First and second beam release

The PCA9685 VCC pin uses ESP32 3.3 V for logic. Its V+ pin uses a separate regulated servo supply. All grounds are joined. If the ultrasonic ECHO signal is 5 V, a 10 k Ω and 15 k Ω divider lowers it before GPIO34. The four motors must not overload the L298N. If their measured combined current is too high, the same ESP32 control signals must drive a higher-current dual-channel motor driver.

Power

The motors use a suitable battery through the motor driver. The ESP32 uses a regulated board input. The five servos use a regulated 5 to 6 V supply through PCA9685 V+. Battery capacity and regulator current must be selected from measured motor and servo current, especially while all four motors start or turn together.

5. Autonomous working and calculations

On power-up, the motors remain off. The ESP32 initializes the PCA9685 at address 0x40, sets the five servo channels to their closed positions, and waits for the GPIO32 button. One press starts the route below.

1. Move forward until the ultrasonic reading is 20 cm or less.
2. Stop, turn 90 degrees left, and open channel 0 for the first two kits.
3. Move until both IR sensors detect the black tape stripe. Stop and open channel 1 for six kits.
4. Continue to the next 20 cm wall reading. Stop and open channel 2 for the other two kits.
5. Turn left, move to the next 20 cm wall reading, and stop.
6. Turn left and open channel 3 for the first beam.
7. Turn left again, open channel 4 for the second beam, and stop permanently.

The robot is autonomous after the button press. The sensors end the forward legs, and a calibrated time controls each turn. The program stops if an ultrasonic echo disappears, a wall is reached before the tape checkpoint, or a movement exceeds its timeout.

Motor speed estimate

The program uses an 8-bit PWM value of 102:

$$\text{duty cycle} = \frac{102}{255} \times 100\% = 40\%.$$

For a motor rated at 500 RPM, $500 \times 0.40 = 200$ RPM is only a starting estimate. Motor speed is not perfectly proportional to PWM, especially with four motors and a loaded chassis. The real wheel speed must be measured.

If wheel diameter is D mm and measured speed is n RPM, forward speed is

$$v = \frac{\pi D n}{60} \text{ mm/s.}$$

Ultrasonic distance and turning

For echo time t in microseconds, the distance is

$$d = \frac{t \times 0.0343}{2} \text{ cm.}$$

The 20 cm threshold is measured from the sensor face. Wheel coasting and the front bracket reduce the actual free space. Left turns currently use 580 ms of opposite wheel motion. That value must be adjusted with the loaded four-wheel chassis because skid steering depends on tyre grip, weight, and battery voltage.

Repository files

The competition upload is `MinimalistCareBot.ino`, stored in the `output/MinimalistCareBot` folder. `CareBotESP32.ino` is a development and fault-checking version. `CareBotHardwareTest.ino` is a bench diagnostic. Uploading either test sketch temporarily replaces the competition program on the ESP32.

References

- [1] IHFC, “Robotics for Good Youth Challenge (RFGYC’26),” [official competition page](#).
- [2] Robotics for Good Youth Challenge, *Rulebook 2026-2027*, supplied competition rulebook.
- [3] IHFC, *Design Document and Video Submission Template*, [reference PDF](#).